
Human–AI Co-Scientist Framework for Quantitative Integration of Multi- Assay Bioactivity Data

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Abstract

Diverse bioactivity assays are widely used to characterize the biological functions of natural products, yet differences in assay principles, measurement scales, and evaluation criteria make integrated interpretation challenging. As a result, multi-assay bioactivity outcomes are often analyzed in a fragmented, assay-specific manner, limiting systematic comparison, reproducibility, and transparency.

In this study, we propose a human–AI co-scientist framework for the quantitative integration of heterogeneous bioactivity data across multiple assays. Within this framework, human researchers retain full control over experimental design, data acquisition, and biological interpretation, while AI systems are employed as analytical partners to support data structuring, standardization, cross-assay comparison, and consistency checking. Bioactivity measurements derived from distinct assays are transformed into a standardized Effect Factor (EF)-based quantitative representation, enabling systematic comparison across assays with differing scales and response characteristics.

Application of the framework to experimentally obtained bioactivity datasets demonstrates that EF-based integration reveals coherent comparative patterns that are difficult to discern from assay-specific analyses alone, while preserving assay-level information. By explicitly structuring the interaction between human expertise and AI-supported analysis, the proposed framework enhances analytical transparency, consistency, and reproducibility without compromising biological interpretability. This work provides a practical methodological reference for integrated multi-assay bioactivity analysis in natural product research and related experimental contexts.

1 Introduction

Diverse bioactivity assays are widely employed to evaluate the biological functions of natural products, including antioxidant, antimicrobial, and anti-inflammatory activities. However, substantial heterogeneity in assay principles, measurement scales, and evaluation criteria poses a major challenge for integrative interpretation and systematic comparison across studies. As a result, bioactivity outcomes are often reported as isolated indicators, limiting their utility for comprehensive evaluation, cross-assay comparison, and higher-level quantitative analysis [1-3].

Despite the widespread adoption of multi-assay bioactivity evaluation, most studies continue to interpret results in an assay-specific or fragmented manner. Heterogeneous bioactivity outcomes are frequently presented side by side without a unified analytical framework, leaving cross-assay comparison largely dependent on qualitative judgment. Although some approaches attempt to aggregate multiple bioactivity indicators, these efforts often rely on ad hoc normalization or partial assay coverage, failing to account for differences in measurement principles, dynamic ranges, and statistical properties. Consequently, integrated bioactivity interpretation remains subjective, difficult to reproduce, and analytically opaque [4-6].

To address these limitations, we propose a human–AI co-scientist framework for the quantitative integration of heterogeneous bioactivity data across multiple assays. In this framework, human

researchers retain full control over experimental design, data acquisition, and biological interpretation, while AI systems function as analytical partners supporting data structuring, normalization, comparative analysis, and consistency checking. By explicitly separating human-driven biological reasoning from AI-supported analytical operations, the framework enables transformation of heterogeneous assay outputs into a comparable quantitative space without relying on assay-specific interpretation. This design supports transparent, reproducible, and interpretable integration of multi-assay bioactivity outcomes. The main contributions of this study are threefold: (i) the introduction of a human–AI co-scientist framework for quantitative multi-assay integration, (ii) demonstration of improved analytical consistency and reproducibility through structured human–AI collaboration, and (iii) provision of a practical methodological reference for unified bioactivity evaluation across diverse experimental settings.

2 Related Work and Background

2.1 Multi-assay bioactivity evaluation in natural product research

Natural product research commonly employs multiple bioactivity assays to capture the diverse biological functions of complex mixtures. Because individual assays probe distinct biochemical mechanisms under different experimental conditions, it has become standard practice to evaluate natural products using combinations of antioxidant, antimicrobial, anti-inflammatory, and related assays. This multi-assay strategy is widely regarded as essential for providing a more comprehensive characterization of biological potential than any single assay alone and has therefore been broadly adopted across natural product chemistry, pharmacognosy, and functional screening studies.

Despite this widespread adoption, the interpretation of multi-assay bioactivity results remains largely assay-specific. Measurements obtained from different assays are typically analyzed and reported independently, and multi-assay results are often presented in parallel without a unified analytical framework for systematic comparison. As a consequence, researchers frequently rely on qualitative judgment when interpreting overall bioactivity trends or prioritizing samples based on heterogeneous assay outcomes, limiting reproducibility and quantitative cross-assay evaluation.

2.2 Limitations of fragmented interpretation and analytical opacity

The increasing availability of multi-assay bioactivity data has not been matched by corresponding advances in integrative analytical practice. In most studies, bioactivity outcomes derived from different assays are analyzed and discussed independently, with limited consideration of their comparability across measurement scales, response distributions, or underlying assay principles. This fragmented approach implicitly assumes that heterogeneous bioactivity indicators can be meaningfully juxtaposed without a unified analytical basis—an assumption that is rarely examined or explicitly justified.

Inconsistencies in data preprocessing, normalization strategies, threshold definitions, and analytical emphasis further complicate meaningful comparison across assays and studies. Even when similar assays are employed, divergent analytical choices can lead to substantially different interpretations from comparable experimental data, making it difficult to distinguish genuine biological variation from methodological artifacts. Moreover, analytical workflows used to transform and integrate bioactivity data are often insufficiently documented, rendering the pathways from raw assay outputs to integrated conclusions opaque and difficult to reproduce. Together, these limitations undermine interpretability, reproducibility, and cross-study comparability, highlighting the need for a more structured and transparent framework for multi-assay bioactivity integration with explicit emphasis on interpretability, human oversight, and reproducibility [7–9]. To address these challenges, we introduce a human–AI co-scientist framework that explicitly structures quantitative integration across heterogeneous bioactivity assays

3 Human-AI Co-Scientist Framework for Integrated Bioactivity Analysis

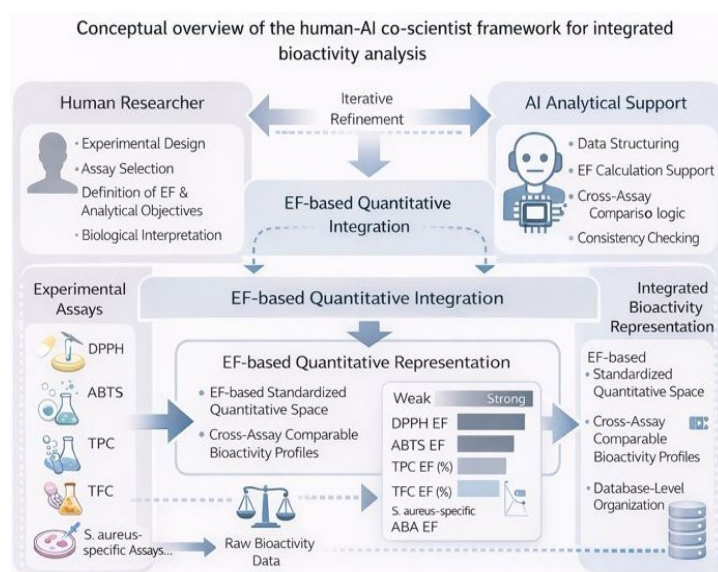


Figure 1. Conceptual overview of the human–AI co-scientist framework illustrating EF-based quantitative integration and cross-assay comparison of heterogeneous bioactivity data within a human-centered and interpretable human–AI analytical framework [7,10].

3.1 Design principles of the human–AI co-scientist framework

The proposed human–AI co-scientist framework is designed to address key limitations in existing multi-assay bioactivity analyses, including fragmented interpretation, analytical opacity, and limited reproducibility. Rather than positioning AI as an autonomous decision-making entity, the framework defines explicit design principles that structure a transparent and complementary interaction between human expertise and AI-supported analysis.

A central principle is the clear separation of biological reasoning and analytical operations. Human researchers retain full responsibility for formulating research questions, selecting bioactivity assays, and interpreting biological meaning, while AI systems support analytical tasks such as data structuring, normalization, comparative assessment, and consistency checking. This separation ensures that interpretative authority remains under human control while analytical procedures are applied in a systematic and traceable manner.

A second principle is quantitative comparability across heterogeneous assays. Because bioactivity assays differ substantially in measurement principles, scales, and statistical properties, the framework emphasizes the transformation of assay-specific outputs into a shared quantitative space. These transformations are explicitly defined rather than heuristic, allowing analytical assumptions to be identified, evaluated, and revised. By prioritizing comparability at the analytical level, the framework reduces reliance on qualitative judgment in multi-assay integration.

Finally, the framework adopts a human-centered approach to AI support. AI systems function as adaptive analytical partners that enhance consistency, efficiency, and exploratory comparison across assays, rather than as black-box predictors. Through these design principles, the framework leverages computational strengths while preserving interpretability, methodological rigor, and scientific accountability.

3.2 Division of analytical roles between human researchers and AI systems

A defining feature of the human–AI co-scientist framework is the explicit division of analytical roles between human researchers and AI systems. This division clarifies responsibility, decision authority, and the scope of computational support, directly addressing interpretative ambiguity in prior multi-assay bioactivity studies.

Human researchers maintain responsibility for all biologically meaningful decisions, including the definition of experimental scope, selection and justification of assays, and interpretation of integrated bioactivity outcomes. They determine which analytical perspectives are relevant and how quantitative results should be contextualized within existing biological knowledge, ensuring that interpretation remains grounded in domain expertise.

AI systems, in contrast, provide structured analytical support focused on data organization, standardized transformation, quantitative comparison across assays, and identification of potential inconsistencies arising during analysis. Importantly, AI does not generate independent biological conclusions or prioritize samples based on opaque criteria. Instead, it implements human-defined analytical objectives in a consistent and reproducible manner.

By explicitly separating interpretative authority from analytical execution, the framework minimizes unintended delegation of biological judgment to algorithmic processes and reduces variability introduced by ad hoc analytical choices. This structured division of labor enables transparent tracing of how integrated bioactivity results emerge from experimental data and analytical operations.

3.3 Quantitative integration workflow for heterogeneous bioactivity data

The framework implements a structured workflow for quantitative integration of heterogeneous bioactivity data derived from multiple assays. The workflow is designed to accommodate differences in assay principles, measurement scales, and statistical characteristics while maintaining transparency at each analytical stage.

The process begins with organization and preprocessing of raw assay data to ensure internal consistency within individual assays. Human researchers define the scope of integration and assess data quality, while AI systems assist in structuring datasets and flagging potential anomalies that may affect downstream analysis.

Assay-specific measurements are then transformed into a comparable quantitative representation using explicitly defined analytical procedures. These transformations preserve relative information within each assay while enabling systematic cross-assay comparison. AI systems apply transformations consistently and support comparison across analytical options, while final methodological decisions remain under human control.

Integrated comparison is subsequently performed within a shared quantitative space, enabling evaluation of relative bioactivity patterns across assays without privileging any single measurement. AI-assisted analysis supports efficient comparison and consistency checking, while interpretation of integrated results remains closely coordinated with human biological judgment. This structured workflow reduces reliance on subjective aggregation and supports reproducible integration across assays and studies.

3.4 Transparency, reproducibility, and responsible AI use

Transparency and reproducibility are core requirements of the proposed framework. All analytical steps—including data preprocessing, transformation, normalization, and comparison—are treated as explicit methodological components rather than implicit technical details. This explicitness allows researchers to trace how integrated bioactivity outcomes arise from raw assay data and to evaluate how alternative analytical choices might influence results.

The structured and standardized nature of the workflow supports reproducibility across datasets and experimental contexts, not through rigid uniformity but through shared analytical logic with clearly articulated assumptions. At the same time, the framework emphasizes responsible AI use by constraining AI to an analytical support role. Interpretative authority remains exclusively with human researchers, who define analytical objectives and contextualize quantitative outcomes within biological knowledge. Through this design, the framework promotes ethical, transparent, and

accountable integration of AI into bioactivity analysis [7, 10].

4 Results

4.1 Overview of datasets and experimental scope

This section presents quantitative results obtained from the integration of heterogeneous bioactivity assays within the proposed human–AI co-scientist framework. The analyzed dataset comprises multiple bioactivity measurements derived from diverse experimental assays designed to capture different biological functions of natural product samples.

Because each assay operates under distinct measurement principles and dynamic ranges, results were first examined at the assay-specific level and subsequently analyzed using a standardized EF-based quantitative representation. All analyses are based exclusively on experimentally obtained bioactivity data, and no predictive modeling or extrapolation beyond observed measurements was performed.

4.2 Assay-specific bioactivity patterns

Sample ID	DPPH EF	ABTS EF	TPC EF (%)	TFC EF (%)	<i>S. aureus</i> -specific ABA EF
HM 1	13.18	8.98	45.84	25.35	2.77
HM 2	14.06	9.65	44.83	29.13	2.94
HM 3	0.18	0.26	2.39	1.27	0.00
HM 4	0.31	0.38	3.99	1.69	0.00
HM 5	0.42	0.90	11.68	7.02	0.00
HM 6	1.10	0.87	13.24	12.71	0.00
HM 7	0.56	1.35	11.19	7.49	0.22
HM 8	2.01	3.50	22.02	17.99	1.52
HM 9	2.70	2.33	24.09	31.97	0.22
HM 10	4.15	3.34	30.54	46.77	0.20
HMP 1	0.21	0.47	2.54	0.85	0.00
HMP 2	0.28	0.58	3.63	1.68	2.12
HMP 3	0.40	0.81	3.11	2.54	0.00
HMP 4	0.34	1.06	4.03	4.55	0.00
HMP 5	0.24	0.33	1.91	0.52	1.18
HMP 6	0.27	0.22	2.02	0.72	6.72
HMP 7	0.37	0.61	4.14	1.95	0.00
HMP 8	0.49	0.93	5.03	2.70	0.19
HMP 9	1.07	2.01	12.31	5.20	2.39
HMP 10	1.04	2.66	14.77	6.09	4.51

Figure 2. EF-based bioactivity heatmap of representative herbal materials and herbal prescriptions. Effect Factor (EF) values from multiple bioactivity assays (DPPH, ABTS, TPC, TFC, and *S. aureus*-specific ABA) are shown for a selected subset of samples from the full dataset (n = 586), illustrating heterogeneous bioactivity distributions across assays.

Assay-specific analysis revealed substantial heterogeneity in bioactivity outcomes across the evaluated experimental assays. Because each assay probes distinct biological mechanisms and operates under different measurement principles, observed bioactivity values exhibited markedly different distributions, dynamic ranges, and relative trends across samples.

When examined independently, relative sample rankings varied considerably across assays. In several cases, samples showing high activity in one assay exhibited moderate or low activity in others, indicating that assay-dependent responses capture complementary but non-equivalent aspects of biological function. As a result, direct comparison of bioactivity magnitudes across assays based on raw outputs remained ambiguous.

Importantly, this variability extended beyond absolute measurement scale to inconsistencies in comparative interpretation. Depending on the assay considered, the same sample could be characterized as highly active, moderately active, or indistinguishable from others. These results demonstrate that while assay-specific analyses provide valuable mechanistic insights, they are insufficient on their own for systematic cross-assay comparison.

4.3 Quantitative integration and cross-assay relationships

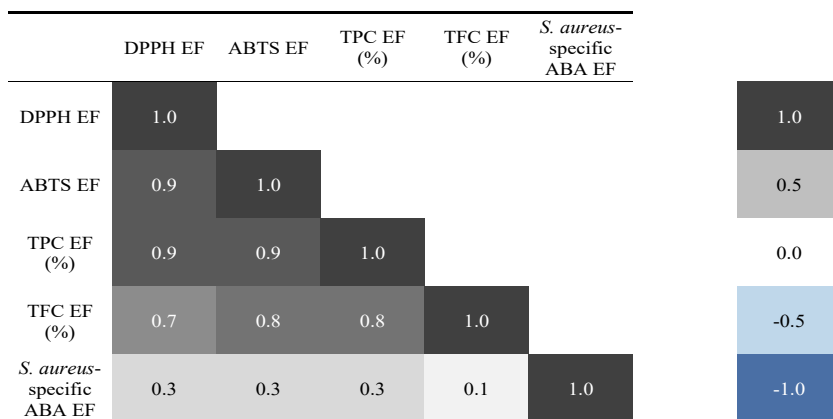


Figure 3. EF-based correlation heatmap of antioxidant and antimicrobial bioactivity assays. Pearson correlation coefficients were calculated using EF values derived from DPPH, ABTS, TPC, TFC, and *S. aureus*-specific antibacterial assays.

To enable systematic comparison across heterogeneous assays, assay-specific bioactivity measurements were transformed into a standardized EF-based quantitative representation that places all results within a common comparative space. This transformation allowed bioactivity outcomes derived from assays with distinct measurement scales and response characteristics to be evaluated on a unified quantitative basis.

Following standardization, previously fragmented assay-specific patterns converged into more coherent comparative structures. Samples that exhibited inconsistent rankings across individual assays became distinguishable based on their integrated bioactivity profiles, revealing relative differences that were not apparent from raw assay outputs alone. Importantly, this integration aligned heterogeneous measurements without eliminating assay-specific information.

4.4 Consistency and robustness of integrated bioactivity results

The integrated EF-based representation exhibited stable comparative patterns across assays and samples. Relative sample positioning within the integrated quantitative space remained largely consistent across different analytical perspectives, indicating that observed trends were not driven by any single assay.

Whereas assay-specific outcomes were strongly influenced by differences in measurement scale and response range, EF-based integration mitigated these effects, allowing coherent comparison across assays with divergent characteristics. Assay-specific information was preserved while contributing proportionally to integrated profiles, preventing dominance by extreme values or highly sensitive assays.

4.5 Summary of key findings

Table 1. Bioactivity assays and EF definitions used in this study.

Bioactivity domain	Assay	Measurement principle	Raw output	EF representation	Dimension
Antioxidant	DPPH	Radical scavenging activity	IC ₅₀ or inhibition (%)	DPPH EF	Dimensionless
Antioxidant	ABTS	Radical cation decolorization	IC ₅₀ or inhibition (%)	ABTS EF	Dimensionless
Antioxidant-related	TPC	Total phenolic content	mg GAE/g	TPC EF	Dimensionless (%)
Antioxidant-related	TFC	Total flavonoid content	mg QE/g	TFC EF	Dimensionless (%)
Antibacterial	<i>S. aureus</i> -specific ABA	Growth inhibition	MIC / inhibition	ABA EF	Dimensionless

Independent assay analyses revealed substantial heterogeneity in bioactivity outcomes, resulting in inconsistent sample rankings across assays. EF-based standardization successfully aligned heterogeneous bioactivity measurements within a shared quantitative space, enabling coherent cross-assay comparison.

Integrated results exhibited stable and reproducible patterns that were not apparent from assay-specific analyses alone. Collectively, these findings establish a quantitative foundation for integrated multi-assay bioactivity analysis, which is interpreted and contextualized in the following Discussion. These integrated patterns form the basis for the interpretative and methodological discussion presented below.

5 Discussion

5.1 Interpretation of integrated bioactivity patterns

The integrated bioactivity analysis provides a coherent interpretative perspective that is difficult to obtain from assay-specific results alone. While individual assays yielded divergent rankings and inconsistent comparative outcomes, EF-based integration revealed stable bioactivity tendencies at the level of overall biological response rather than isolated assay performance.

Importantly, integrated patterns contextualize—rather than replace—assay-specific insights. Samples exhibiting consistent activity across multiple assays could be distinguished from those showing strong responses in only a single assay, enabling a more nuanced assessment of biological relevance. This distinction is particularly meaningful in natural product research, where complex mixtures often elicit multifaceted effects that cannot be captured by a single experimental readout [11, 12].

5.2 Methodological implications of EF-based standardization

EF-based standardization addresses a central limitation of multi-assay bioactivity analysis by enabling direct quantitative comparison across assays with heterogeneous measurement scales and response characteristics. By transforming assay-specific outputs into a shared comparative space, EF reduces ambiguity arising from scale-dependent variability and supports more transparent integrative interpretation.

Notably, EF does not impose uniform weighting or oversimplify assay-specific information. Instead, it preserves the relative contribution of individual assays while enabling systematic comparison across bioactivity dimensions. This balance between standardization and interpretability supports the use of EF as a general analytical strategy for multi-assay integration rather than as an assay-specific metric.

5.3 Role of the human–AI co-scientist framework

The human–AI co-scientist framework provided a structured environment for implementing EF-based integration in a transparent and reproducible manner. Human researchers retained full responsibility for experimental design and biological interpretation, while AI systems supported data structuring, standardization, and consistency checking across analytical steps.

This explicit division of roles mitigates risks associated with opaque automation by ensuring that interpretative authority remains with human expertise. At the same time, AI-assisted analysis enhanced analytical consistency and reduced the cognitive burden associated with handling complex multi-assay datasets, demonstrating the practical value of a collaborative human–AI paradigm in quantitative bioactivity analysis.

5.4 Methodological considerations and limitations

Several limitations should be considered when interpreting the integrated results. Integrated bioactivity patterns are inherently constrained by the selection of included assays and reflect only the biological dimensions captured by those measurements. Moreover, while EF-based standardization mitigates scale-dependent variability, it does not eliminate all sources of experimental or analytical noise.

Additionally, integration necessarily involves abstraction, which may obscure subtle assay-specific effects relevant in certain biological contexts. Accordingly, integrated analysis should be regarded as a complementary analytical layer rather than a replacement for detailed assay-level investigation.

5.5 Broader applicability and future directions

The proposed framework is readily extensible to other multi-assay screening contexts in natural product research and related fields. Future work may incorporate expanded assay panels, larger datasets, or context-aware extensions of EF-based standardization to enable adaptive weighting of assay contributions. More broadly, the human–AI co-scientist paradigm demonstrated here offers a generalizable model for responsible integration of AI into experimental data analysis [11, 12].

6 Conclusion

This study presents a human–AI co-scientist framework for the quantitative integration of heterogeneous bioactivity data obtained from multiple experimental assays. By combining EF-based standardization with structured human–AI collaboration, the proposed approach enables systematic cross-assay comparison while preserving assay-specific biological information.

The results demonstrate that integrated bioactivity analysis reveals coherent and robust patterns that are difficult to discern from individual assays alone. By addressing scale-dependent variability and fragmented interpretation, the framework provides a transparent and reproducible foundation for multi-assay bioactivity evaluation.

Overall, this work establishes a practical methodological basis for integrated bioactivity analysis in natural product research and related domains, supporting future extensions toward larger datasets, expanded assay panels, and responsible AI-assisted scientific workflows.

Acknowledgments and Disclosure of Funding

The author declares that this research was conducted without any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The author also declares no competing financial or non-financial interests related to this work.

References

- [1] Cos, P., Vlietinck, A.J., Berghe, D.V. & Maes, L. (2006) Anti-infective potential of natural products: How to develop a stronger in vitro “proof-of-concept”. *Journal of Ethnopharmacology* 106(3): 290–302.
- [2] Atanasov, A.G., Waltenberger, B., Pferschy-Wenzig, E.-M., et al. (2015) Discovery and resupply of pharmacologically active plant-derived natural products: A review. *Biotechnology Advances* 33(8): 1582–1614.
- [3] Harvey, A.L., Edrada-Ebel, R. & Quinn, R.J. (2015) The re-emergence of natural products for drug discovery in the genomics era. *Nature Reviews Drug Discovery* 14(2): 111–129.
- [4] van den Berg, R.A., Hoefsloot, H.C.J., Westerhuis, J.A., Smilde, A.K. & van der Werf, M.J. (2006) Centering, scaling, and transformations: Improving the biological information content of metabolomics data. *BMC Genomics* 7: 142.
- [5] Broadhurst, D. & Kell, D.B. (2006) Statistical strategies for avoiding false discoveries in metabolomics and related experiments. *Metabolomics* 2(4): 171–196.
- [6] Westerhuis, J.A., Hoefsloot, H.C.J., Smit, S., et al. (2008) Assessment of PLSDA cross validation. *Metabolomics* 4(1): 81–89.
- [7] Doshi-Velez, F. & Kim, B. (2017) Towards a rigorous science of interpretable machine learning. *arXiv preprint* arXiv:1702.08608.
- [8] Amershi, S., Weld, D., Vorvoreanu, M., et al. (2019) Guidelines for human–AI interaction. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, Paper No. 3.
- [9] Pineau, J., Vincent-Lamarre, P., Sinha, K., et al. (2021) Improving reproducibility in machine learning research (A report from the NeurIPS 2019 reproducibility program). *Journal of Machine Learning Research* 22(164): 1–20.
- [10] Lipton, Z.C. (2018) The mythos of model interpretability. *Communications of the ACM* 61(10): 36–43.
- [11] Wishart, D.S. (2016) Emerging applications of metabolomics in drug discovery and precision medicine. *Nature Reviews Drug Discovery* 15(7): 473–484.
- [12] Kell, D.B. & Oliver, S.G. (2016) The metabolome 18 years on: A concept comes of age. *Metabolomics* 12: 148.

AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: [The abstract and introduction clearly state the paper's contribution as a human-AI co-scientist framework for EF-based quantitative integration and cross-assay comparison of heterogeneous bioactivity data, without claiming predictive modeling or out-of-dataset generalization (Abstract; Introduction). The scope of the claims is consistent with the EF-based visualization and correlation analyses presented in Sections 4.2–4.3.]

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: [The limitations of the proposed framework, including assay dependence, abstraction effects of EF-based integration, and lack of predictive modeling, are explicitly discussed in Section 5.4 (Methodological considerations and limitations).]

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: [All experimental results are based on experimentally obtained bioactivity data, and the analytical workflow—including EF calculation, standardization, and cross-assay comparison—is described in detail in Section 3.3 (Quantitative integration workflow) and Sections 4.1–4.3, enabling reproduction of the reported results.]

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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based bioactivity patterns (Sections 4.2–4.3), and therefore does not report error bars or significance tests.]

8. **Experiments Compute Resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

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Justification: [The analysis does not involve computationally intensive model training or large-scale optimization. EF calculation and integration were performed using standard analytical tools, and detailed compute resource reporting is not applicable.]

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